

Ultrafast Resonance-Tracking Thin-Film Lithium Tantalate Resonator: Enabling Colorless Self-Coherent Reception

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Abstract We demonstrate a high-Q thin-film lithium tantalate (TFLT) microring resonator featuring a fast adaptive resonance-tracking speed with a convergence time below $1.3 \mu\text{s}$. The ultrafast-tracking TFLT resonator enables colorless carrier-extracted self-coherent reception, achieving 375-Gbps transmission over 50-km SSMF using an uncooled DFB laser. ©2026 The Author(s)

Introduction

The exponential growth in data-centre interconnects and high-capacity optical communication links has fuelled an insatiable demand for compact, power-efficient, and versatile photonic integrated circuits (PICs) [1], [2]. Within such PICs, microring resonators (MRRs) are fundamental building blocks for wavelength filtering, modulation, and signal processing [3]. However, their practical deployment in dynamic environments has been hindered by a critical bottleneck: environmental fluctuations and thermal drift cause severe resonance misalignment, degrading device performance and system stability.

Conventionally, active resonance locking is employed to stabilize MRRs against such perturbations [4]. The tracking speed of most feedback systems is limited by the thermo-optic tuning mechanism, typically restricting compensation to the microsecond range [5], [6], [7], [8]. This low-speed tracking is insufficient for burst-mode traffic, rapidly changing environmental conditions, or scenarios requiring fast wavelength agility, thereby limiting the applicability of MRRs in future agile and reconfigurable networks. To overcome the speed limitations, electro-optic (EO) tuning presents a promising alternative. Among EO materials, lithium niobate (LN) has demonstrated ultrafast-tracking adaptive polarization control in prior research [9]. Nevertheless, lithium tantalate (LT), which shares similarly superior EO properties with LN [10], offers the distinct advantage of substantially lower direct-current (DC) drift [11], [12], rendering it particularly suitable for stable and high-speed tracking operations.

In this work, we demonstrate a thin-film lithium tantalate (TFLT) MRR with an ultrafast adaptive resonance-locking speed ($1.06\text{-}\mu\text{s}$ tracking-

response convergence time). Specifically, we harness this high-speed resonator to enable a novel colorless carrier-extracted self-coherent (CESC) reception scheme [13], [14] as shown in Fig. 1 (a). By utilizing an uncooled distributed feedback (DFB) laser, whose inherent wavelength drift is now instantaneously compensated by the ultrafast-tracking filter, we successfully demonstrate a 375-Gbps transmission over 50-km standard single-mode fibre (SSMF). The ultrafast resonance tracking speed and the capability to adaptively compensate the wavelength drift of an uncooled DFB laser, thereby enabling successful self-coherent transmission, validate the lithium tantalate on insulator (LTOI) platform as a key enabler for high-capacity, low-cost, and colorless self-coherent receptions.

TFLT MRR with Fast Resonance Tracking

Fig. 1(a) presents a conceptual schematic of the heterogeneously integrated LO-free CSEC transceiver, where the ultrafast-tracking high-Q TFLT MRR in this work is well-suited for applications. This ideal integration can combine the superior EO properties of the LTOI platform [15] with the mature passive components and photodiodes of the silicon photonics (SiPh) platform [16]. Extracting the optical carrier from the self-coherent signal where the modulated signal and carrier are co-transmitted, the CESC receiver enables coherent-like detection without LOs. The ultrafast-tracking MRR relaxes laser requirements and enables colorless self-coherent reception, which is the key focus of this work. Fig. 1(b) shows a microscope image of the LTOI MRR, with EO-tuning electrodes distributed along both sides of the waveguide. As depicted in Fig. 1(c), the LT MRR exhibits a 0.685-nm free-spectral range (FSR),

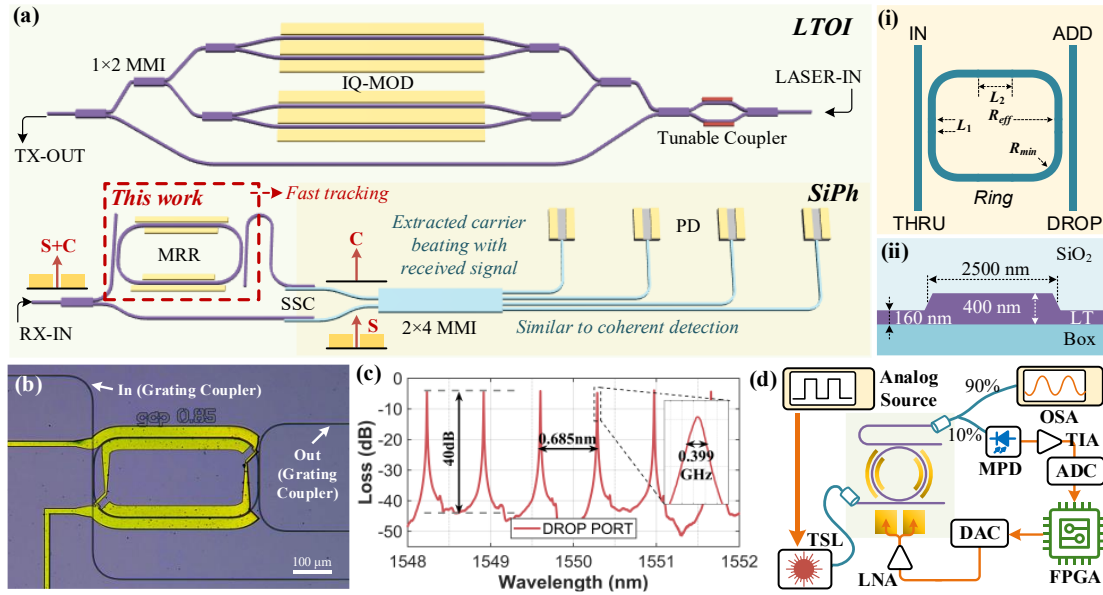


Fig. 1: (a) Conceptual schematic of a carrier-extracted self-coherent (CESC) transceiver based on heterogeneous integration of a LT IQ modulator, a fast-tracking LT MRR, and a SiPh receiver. (b) Microscope image of the LT MRR filter. (c) Spectral response of the filter's drop port. (d) Experimental setup for LTOI MRR tracking. Insets: (i) Schematic of LT MRR structure. (ii) Cross-section diagram of the LT waveguide. SSC: spot-size convertor. TSL: tunable semiconductor laser. MPD: monitoring photodiode. OSA: optical spectrum analyzer. LNA: low-noise amplifier. FPGA: field programmable gate array. ADC: analog - to-digital convertor. DAC: digital-to-analog convertor. $L_1 = 35 \mu\text{m}$, $L_2 = 370 \mu\text{m}$, $R_{\text{eff}} = 120 \mu\text{m}$, $R_{\text{min}} = 84.731 \mu\text{m}$.

40-dB extinction ratio, and 0.399-GHz 3-dB bandwidth, corresponding to a Q-factor of approximately 484500.

Fig. 1(d) depicts the wavelength locking setup. Light from the laser couples into the LTOI resonator, whose drop power (10%) is detected by an InGaAs PD, then amplified and digitized by a 16-bit ADC. The digitized signal is processed by an FPGA implementing a joint scan-lock control algorithm. During initialization, a first-in-first-out (FIFO)-based sliding window is used to update

the peak-power reference coarsely. The scan-to-lock transition is then determined relative to this reference, improving robustness against noise and transient fluctuations. In the subsequent scan mode, it rapidly tunes the MRR to approach the target wavelength; upon reaching 90% of the ideal intensity, it switches to lock mode, where a PID controller adjusts the control voltage via a 14-bit DAC to precisely align resonance.

Fig. 2(a) validates the locking performance under a source wavelength switching from 1550 nm to 1570 nm. Under different wavelength switching ranges, the LT MRR resonance can be fast and stably locked, presenting its good wavelength locking performance over a wide wavelength range. As shown in Fig. 2(b), the drive voltage rapidly follows and remains stable to maintain resonance locking when the source wavelength is stepped from 1550 nm to 1550.5 nm (within a FSR) in 0.1-nm increments. Fig. 3 demonstrates ultrafast tracking response using a square-wave-modulated source. The MRR tracking responds in 1.3 μs , while the rise time of the source response is 0.24 μs . The system achieves a 1.06- μs (1.3 μs - 0.24 μs) tracking response convergence time which is normalized by laser modulation response, significantly outperforming conventional thermo-optic tuning-based locking schemes.

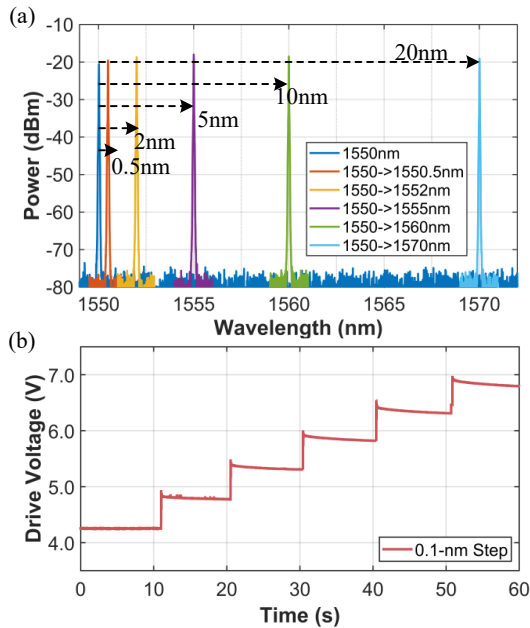


Fig. 2: (a) Optical spectra of locked MRR resonances with different source wavelength switching from 1550nm. (b) Measured drive voltage (DAC output before LNA) under 0.1-nm-step source wavelength shifting.

System Validation

Fig. 4(a) illustrates the experimental setup used to validate the wavelength-tracking performance of the LTOI resonator in a single-polarization

CESC system with an uncooled DFB laser as the light source. The laser output is split into two paths: one feeds an IQ modulator driven by a 120-GSa/s arbitrary waveform generator (AWG, Keysight M8194A) to generate the signal, while the other serves as the optical carrier. The modulated signal and carrier are then combined for transmission. Two OFDM signals are transmitted: a 300-Gb/s 16-QAM signal and a 375-Gb/s 32-QAM signal. A 3-GHz guard band is inserted at the middle of the OFDM signal to facilitate clean optical carrier extraction. At the receiver, the incoming signal is pre-amplified by an EDFA and split into two branches. One branch passes through the LT MRR for optical carrier extraction; the other remains as the signal path. Both branches are fed into an optical hybrid for same-source coherent detection. The output RF signals are captured by a 256-GSa/s real-time oscilloscope (RTO, Keysight UXR0594AP) for offline DSP. An additional EDFA is placed after carrier extraction to compensate for chip-coupling losses (~12 dB) incurred by the grating couplers. This EDFA can be eliminated by adopting low-loss edge couplers or fully integrating the receiver [13], [17].

Fig. 4(b) shows the BER performance versus OSNR (excluding carrier power) for the LT MRR-based self-coherent system under OBTB and after 50-km SSMF transmission cases. Considering the 20% SD-FEC threshold (2.4×10^{-2}) [13], the receiver sensitivities for 300 Gb/s and 375 Gb/s are approximately 13.5 dB and 19 dB, respectively. Fig. 4(c) further demonstrates that the ultrafast-tracking filter enables stable operation across different source wavelengths, with minor performance variations primarily arising from grating coupler bandwidth limitations. These transmission results confirm that the ultrafast-tracking TFLT MRR supports colorless self-coherent reception with robust wavelength tracking, even for uncooled DFB lasers, offering a practical

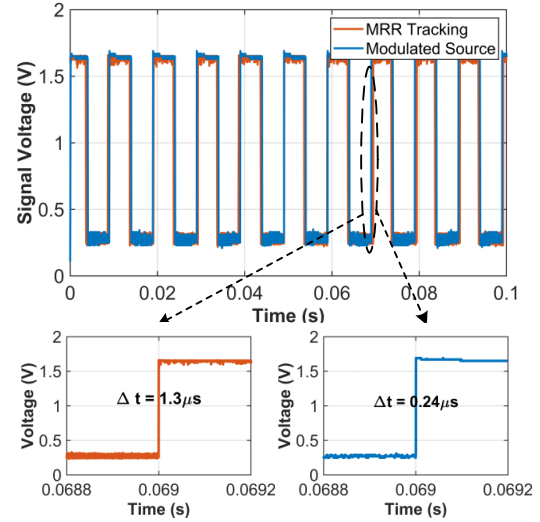


Fig. 3: Tracking waveform detected by the MPD while the optical source is square-wave modulated. (The voltages are normalized to the same scale.)

path toward deploying self-coherent detection in optical interconnects and access networks.

Conclusions

In this paper, we have demonstrated an ultrafast resonance tracking speed (1.06- μ s convergence time) for MRRs based on the LTOI platform. Using an uncooled DFB laser as the optical source, the TFLT-MRR-based CESC system achieves 375-Gb/s transmission over 50-km SSMF while robustly tracking source wavelength drift, further validating its tracking speed and robustness. Ensures rapid resonance locking, this fast-tracking-MRR-based self-coherent scheme is particularly well-suited for access network scenarios, where colorless upstream transmission, fast burst-mode acquisition, and low-cost transceivers are essential. This work offers a performant and potentially low-cost solution for future 100G/200G PON systems and optical interconnects.

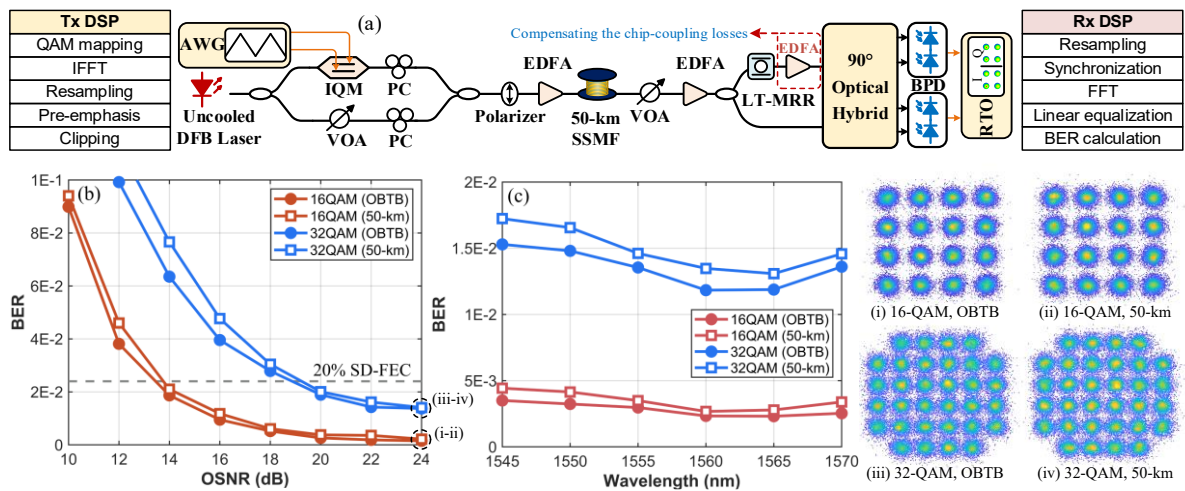


Fig. 4: (a) Experimental setup for system-level validation of the LTOI CROW filter. (b) BER performance versus the OSNR of the received signal. (c) BER performance versus the laser source wavelength. Insets: (i)-(iv) Recovered constellations of transmitted signals. IQM: IQ modulator. PC: polarization controller. VOA: variable optical attenuator. EDFA: erbium-doped fiber amplifier. BPD: balanced photodiode. BER: bit error rate. OSNR: optical signal-to-noise ratio.

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